

Maternal Exposure to Per- and Polyfluoroalkyl Substances (PFAS) and Male Reproductive Function in Young Adulthood: Combined Exposure to Seven PFAS

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BACKGROUND: Concerns remain about the human reproductive toxicity of the widespread per- and polyfluoroalkyl substances (PFAS) during early stages of development.

OBJECTIVES: We examined associations between maternal plasma PFAS levels during early pregnancy and male offspring reproductive function in adulthood.

METHODS: The study included 864 young men (age range: 18.9–21.2 y) from the Fetal Programming of Semen Quality (FEPOS) cohort established between 2017 and 2019. Plasma samples from their mothers, primarily from the first trimester, were retrieved from the Danish National Biobank and levels of 15 PFAS were measured. Seven PFAS had detectable levels above the limit of detection in >80% of the samples and were included in analyses. Semen quality, testicular volume, and levels of reproductive hormones and PFAS were assessed in the young men. We used weighted quantile sum (WQS) regression to estimate the associations between combined exposure to maternal PFAS and reproductive function, and negative binomial regression to estimate the associations of single substances, while adjusting for a range of *a priori*–defined fetal and postnatal risk factors.

RESULTS: By a 1-unit increase in the WQS index, combined maternal PFAS exposure was associated with lower sperm concentration (–8%; 95% CI: –16%, –1%), total sperm count (–10%; 95% CI: –17%, –2%), and a higher proportion of nonprogressive and immotile sperm (5%; 95% CI: 1%, 8%) in the young men. Different PFAS contributed to the associations with varying strengths; however, perfluoroheptanoic acid was identified as the main contributor in the analyses of all three outcomes despite the low concentration. We saw no clear association between exposure to maternal PFAS and testicular volume or reproductive hormones.

DISCUSSION: In a sample of young men from the general Danish population, we observed consistent inverse associations between exposure to maternal PFAS and semen quality. The study needs to be replicated in other populations, taking combined exposure, as well as emerging short-chain PFAS, into consideration. <https://doi.org/10.1289/EHP10285>

Introduction

Per- and polyfluoroalkyl substances (PFAS) have been in use since the 1940s and now constitute a group of more than 5,000 manmade chemicals.^{1,2} Their water-, dirt-, and oil-repellent properties favor their addition to multiple industrial and commercial applications, such as food packaging, coated cookware, cosmetics, textiles, carpets, paints, lubricants, and firefighting foams.^{1–5} In humans, consumption of contaminated food and drinking water is the primary source of systemic uptake followed by inhalation and dermal absorption.^{6,7} The legacy PFAS, such as perfluorooctanoic acid (PFOA), perfluorooctane sulfonic acid (PFOS), perfluorononanoic

acid (PFNA), and perfluorodecanoic acid (PFDA), have long elimination half-lives of 3–8 y^{1,8} and may, therefore, accumulate in the human body. The half-lives are largely dependent on the carbon chain length of the specific PFAS, with long carbon length PFAS having the longest half-life.^{8,9} A few PFAS are regulated in the European Union and internationally. PFOA and PFOS have been listed under the Stockholm Convention on Persistent Organic Pollutants (POPs),¹⁰ and additional PFAS have been classified as “substances of very high concern” by the EU commission [Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)] owing to their bioaccumulative, environmentally persistent, and toxic properties.^{5,11–13} Historically, PFOA and PFOS have been the two most used and also most studied PFAS.¹⁴ Although these substances have been gradually phased out of production in several, mainly developed, countries, they are still detectable in serum worldwide.^{14,15} The use of and exposure to most legacy PFAS are declining¹⁶ and affecting industrial development and the use of new congeners,^{17,18} which may be just as persistent and toxic as the legacy PFAS.¹⁷ In Sweden, where the use of PFAS is similar to that in Denmark, perfluoroheptanoic acid (PFHpA) has not been found to a large extent in food items, which could explain the very low concentrations in adolescents.¹⁹ A new Swedish study found a decrease in concentrations of PFOA, PFOS, and perfluorohexane sulfonic acid (PFHxS) among Swedish adolescents between 2000 and 2017, indicating that important exposure sources have been eliminated.¹⁹ Temporal trends are less clear for other substances, such as PFHpA, PFNA, and PFDA, which might be due to very low concentrations.¹⁹ However, decreasing trends of PFNA and PFDA concentrations were found between 2009 and 2017¹⁹ and

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Supplemental Material is available online (<https://doi.org/10.1289/EHP10285>).

A.G. has received research grants from Ferring Pharmaceuticals and personal fees from Besins Healthcare Nordic and Sandoz, unrelated to the submitted work. The other authors declare they have no actual or potential competing financial interests.

Received 9 September 2021; Revised 22 August 2022; Accepted 26 August 2022; Published 5 October 2022.

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have been consistently rediscovered in studies published after 2010.^{20,21} In Denmark, there has been a decrease in the concentrations of PFAS found among young men reporting for the military draft.^{22,23} However, temporal trends may differ across countries and regions, especially with regard to the continuing use of legacy PFAS,¹⁶ and thus, there are still areas with high levels.^{10,24} PFAS possess endocrine-disrupting properties,^{20,25,26} and animal studies have shown that exposure to PFAS may be associated with decreases in serum testosterone levels, and epididymal sperm counts^{2,27} and altered thyroid hormone homeostasis that may potentially affect fertility.²⁸ Human exposure to PFAS has been linked to an increased risk of cancer²⁹ and adverse effects on the metabolic, endocrine, immune, and reproductive systems,³⁰ but the literature on associations between exposure to PFAS and male reproductive function is so far inconclusive.^{18,25,31,32} In a recent review, we concluded that the amount of data available on male reproductive health risks following PFAS exposures prior to adulthood is limited, and concerns regarding reproductive toxicity, especially in the early stages of fetal development, remain.¹⁸ Studies have shown that several PFAS can cross the placental barrier and accumulate in fetal tissues.^{4,25,26,33,34} One previous epidemiological study found no association between cord blood levels of PFOA and PFOS and congenital cryptorchidism. This might be due to a lack of power.³⁵ Another study, using the Puberty Cohort nested within the Danish National Birth Cohort (DNBC), found that fetal exposure to some PFAS was associated with lower mean ages of male puberty, whereas other PFAS were associated with higher mean ages of male puberty.³⁶ This might be due to disturbance of the hypothalamic–pituitary–gonadal axis, which also plays an essential role in gonad development and spermatogenesis.^{36,37} To date, only three studies have examined fetal exposure to PFOA and PFOS and reproductive parameters such as semen quality and reproductive hormones.^{38–40} One of these studies observed trends of lower sperm concentration, total sperm count, and higher levels of follicle-stimulating hormone (FSH) and luteinizing hormone (LH) with increasing fetal exposure to PFOA.³⁸ The two other studies addressed reproductive hormone levels. They reported an association between fetal exposure to PFOS, measured in amniotic fluid⁴⁰ or maternal plasma at birth,³⁹ and increased levels of testosterone in amniotic fluid⁴⁰ and estradiol plasma levels in male infants,³⁹ respectively. These earlier studies only addressed PFOA and PFOS and not other legacy substances or emerging PFAS.² They did not consider the combined maternal PFAS exposure¹ or the possible contributions from PFAS exposure in adulthood. Moreover, none of the studies measured PFAS exposure during early pregnancy, which is the primary developmental period for the male reproductive organs.⁴¹

The aim of the present study was, therefore, to assess associations between combined⁴² and single-substance maternal PFAS concentrations primarily during early pregnancy and male reproductive function in young adulthood while taking adulthood PFAS exposures into account.

Methods

Study Population

We used data from the Fetal Programming of Semen Quality (FEPOS) cohort,⁴³ which consists of sons born to mothers who, while pregnant, had participated in the DNBC between 1998 and 2003.⁴⁴ A more detailed description of the FEPOS cohort has previously been published.⁴³ From March 2017 to December 2019, we invited sons who were a minimum of 18 years and 9 months of age and living in relative proximity to one of the two study centers in Copenhagen and Aarhus to participate. Sons were considered eligible if their mothers had provided a blood sample during pregnancy and responded to the two DNBC computer-assisted telephone interviews around gestational weeks 16 and 30. Criteria for exclusion were a history of cancer treatment, sterilization, or orchiectomy procedures. Upon signing a consent form, the young men answered a comprehensive electronic questionnaire, provided a semen sample and a blood sample, and underwent a clinical examination at one of the study centers. A total of 5,697 eligible young men were invited, and of these, 1,058 (19%) participated in the FEPOS cohort.⁴³ Of these 1,058, we excluded those with an insufficient amount of maternal plasma available for chemical analyses ($n = 188$) and those who did not have both testicles descended in the scrotum ($n = 6$). Thus, the final study population consisted of 864 men.

Assessment of PFAS Exposure

Maternal plasma from 1998 to 2003 (100 μ L used for analysis) was primarily collected during the first trimester ($n = 821$, 95%), and 5% ($n < 45$) were collected during the second trimester and <1% ($n < 5$) during the third trimester. Maternal plasma samples were stored at -80°C in the Danish National Biobank. A blood sample was collected at the clinical examination from all sons included in this study, as part of the FEPOS data collection between 2017 and 2020, and stored at -80°C until analyses. In 2020 and 2021, 15 different PFAS were measured in plasma from both mother (Table 1) and son⁴⁶ using triple quadrupole linear ion trap mass spectrometers equipped with TurboIonSpray sources (QTRAP 5500; AB Sciex) coupled to a liquid chromatography/tandem mass spectrometry (LC-MS/MS) system (UFLCXR; Shimadzu Corp.).¹⁹ Analyses were performed in two batches between 2019 and 2021 at

Table 1. Limits of detection for and distributions of PFAS plasma concentrations among the 864 mothers of sons included in this study ($n = 864$).

Abbreviation	Chemical name	Carbon chain length	LOD (ng/mL)			Concentration (ng/mL) ^a						
			Batch 1	Batch 2	N (%) > LOD	5th P	25th P	Median	75th P	95th P	Maximum	IQR
PFHxS	Perfluorohexane sulfonic acid	6	0.02	0.04	864 (100)	0.33	0.56	0.77	1.02	1.76	15.72	0.46
PFHpA	Perfluoroheptanoic acid	7	0.02	0.01	765 (88.5)	0.01	0.03	0.06	0.11	0.28	0.61	0.08
PFOA	Perfluorooctanoic acid	8	0.02	0.03	864 (100)	2.04	3.24	4.40	5.87	8.23	16.63	2.63
PFOS	Perfluorooctane sulfonic acid	8	0.02	0.05	864 (100)	14.75	21.68	27.56	34.45	47.49	92.59	12.74
PFNA	Perfluorononanoic acid	9	0.02	0.01	864 (100)	0.20	0.30	0.38	0.49	0.70	2.21	0.19
PFDA	Perfluorodecanoic acid	10	0.02	0.01	864 (100)	0.08	0.12	0.15	0.19	0.28	0.63	0.07
PFUnDA	Perfluoroundecanoic acid	11	0.02	0.01	864 (100)	0.06	0.09	0.12	0.16	0.23	0.36	0.06

Note: Maternal plasma samples were collected during pregnancy, primarily in first trimester and analyzed in two batches. Other measured PFAS not detectable above the LOD in >80% of samples, LOD in ng/mL: PFPeA (0.08), PFHxA (0.07), PFDoA (0.2), PFTTrDA (0.1), PFBS (0.05), PFPeS (0.02), PFHpS (0.1), and PFDS (0.1).⁴⁵ IQR, interquartile range; LOD, limit of detection; P, percentile; PFAS, per- and polyfluoroalkyl substances; PFBS, perfluorobutanesulfonic acid; PFDA, perfluorodecanoic acid; PFDoA, perfluorododecanoic acid; PFDS, perfluorodecane sulfonic acid; PFHpA, perfluoroheptanoic acid; PFHpS, perfluoroheptanesulfonic acid; PFHxA, perfluorohexanoic acid; PFHxS, perfluorohexane sulfonic acid; PFNA, perfluorononanoic acid; PFOA, perfluorooctanoic acid; PFOS, perfluorooctane sulfonic acid; PFPeA, perfluoropentanoic acid; PFPeS, perfluoropentanesulfonic acid; PFTTrDA, perfluorotridecanoic acid; PFUnDA, perfluoroundecanoic acid.

^aPseudo median and percentiles were based on information from at least five individuals with values closest to the actual median/percentile.

the Division of Occupational and Environmental Medicine at Lund University, Sweden. This laboratory is part of the Erlangen Round Robin interlaboratory control program and has qualified as a European Human Biomonitoring Initiative (HBM4EU) laboratory for the analysis of PFAS. Both sample batches included chemical blanks, two homemade quality control (QC) samples and two QC samples with standard reference material. Coefficients of variation (CVs) were between 2% and 13%, and the performance of analyzed standard reference materials are shown in Tables S1 and S2. Limits of detection (LODs) for the individual batches and compounds are shown in Table 1. Seven of 15 PFAS were detectable above the LOD in >80% of the maternal samples and were included in this study: PFHxS, PFHpA, PFOA, PFOS, PFNA, PFDA, and perfluoroundecanoic acid (PFUnDA). All values below the respective LODs were assigned the value of the LOD divided by the square root of 2. The PFAS that were not detectable above the LOD in >80% of the samples were perfluorobutanesulfonic acid (PFBS), perfluoropentanoic acid (PFPeA), perfluoropentanesulfonic acid (PFPeS), perfluorohexanoic acid (PFHxA), perfluoroheptanesulfonic acid (PFHpS), perfluorodecane sulfonic acid (PFDS), perfluorododecanoic acid (PFDoA), and perfluorotridecanoic acid (PFTTrDA).

Semen Analyses

Participants were asked to be sexually abstinent for 48–72 h prior to collection of the semen sample. The actual length of abstinence was recorded at the clinical examination and any spillage was also noted. Semen samples were analyzed by two trained biomedical laboratory technicians affiliated with the study center in Copenhagen and Aarhus, respectively. The technicians were unaware of the exposure status of the sons and participated in a follow-up QC program with the Reproductive Medicine Center in Malmö.⁴³ All semen analyses followed the recommendations of the World Health Organization from 2010.⁴⁷ Semen volume was measured by weighing the sample in a preweighed container, after which the semen was analyzed manually for sperm concentration, total sperm count (sperm concentration multiplied by semen volume), and motility. Sperm concentration was determined on two aliquots diluted with sodium bicarbonate solution that were transferred to each chamber of a BLAUBRAND improved Neubauer hemocytometer and rested in a humidified chamber for 10 min, after which the hemocytometer was examined until at least 200 cells had been counted. Standard dilutions (1:50, 1:20, or 1:10) were used where applicable; however, in very few cases, 1:1, 1:2, 1:5, and 1:14 was used. Sperm cell motility was determined as the percentages of progressive, nonprogressive, and immotile sperm among 200 spermatozoa in each of two fresh drops of semen, placed on a preheated (37°C) clean glass slide with a cover slip. In the statistical analyses, we modeled the sum of nonprogressive and immotile sperm because progressive sperm did not give a satisfactory model fit. The percentages of morphologically normal sperm were analyzed at the Center of Reproductive Medicine in Malmö, Sweden.⁴³

Testicular Volume

During the clinical examination, participants measured their own testicular volume in privacy using a Prader Orchidometer. This method has previously been shown to be valid compared with measurement by an experienced examiner. The measure of agreement was 71% of the right testicles ($\kappa=0.67$) and 76% of the left testicles ($\kappa=0.59$).⁴⁸ The average testicular volume of both testicles was used in the analyses.

Levels of Reproductive Hormones

Levels of reproductive hormones were measured in nonfasting plasma samples from the young men. Levels of testosterone and

estradiol were analyzed using high-pressure LC-MS/MS detection (AB Sciex 6500 QTRAP).⁴⁹ For testosterone, the samples were pipetted onto a protein precipitation plate with acetonitrile containing isotopically labeled internal standards. The protein precipitation plate was shaken, and the supernatants were aspirated into a 96-well collection tray, where they were evaporated using nitrogen. After redissolution, the testosterone level was analyzed using LC-MS/MS. For estradiol, samples were diluted with water, a stable isotope-labeled internal standard was added, and each sample was transferred to a Novum extraction plate, where they were extracted with a mixture of heptane and ethyl acetate. The extracts were evaporated using nitrogen, and after redissolution, the estradiol level was analyzed using LC-MS/MS. LODs were 0.12 nmol/L for testosterone and 15 pmol/L for estradiol, and the CVs were 7% at 14.1 nmol/L and 7.5% at 106 pmol/L, respectively. Calculated free testosterone levels were derived using the Vermeulen formula, assuming a constant albumin concentration of 43 g/L.⁵⁰

$$[FT] = \frac{-b + \sqrt{b^2 + 4a[T]}}{2a}$$

$a = K_{at} + K_t + (K_{at} * K_t)([SHBG] + [albumin] - [T])$ and $b = 1 + K_t[SHBG] + K_{at}[albumin] - ((K_{at} + K_t)[T])$. K_{at} is the association constant of albumin binding to T , whereas K_t is the association constant of SHBG binding to T .⁵⁰

Levels of FSH, LH, and sex hormone-binding globulin (SHBG) were measured using immunoassays (Cobas 8000 e602; Roche Diagnostics)⁵¹ with CVs of 3%, 3%, and 5%, respectively. LODs were 0.1 IU/L for FSH and LH, and 0.350 nmol/L for SHBG. All hormones were analyzed at the Department of Clinical Biochemistry at Aarhus University Hospital, Denmark. Values below the LODs were assigned the value of the LOD divided by the square root of 2.

Covariates

Information on potential risk factors linked to both maternal PFAS concentrations and male reproductive function was obtained from the first DNBC telephone interview of the mothers conducted around gestational week 16.⁵² This interview included self-reported information on maternal prepregnancy body mass index (BMI; kg/m²), smoking during the first trimester [nonsmoking, light smoking (≤ 10 cigarettes/d), heavy smoking (> 10 cigarettes/d)], and household occupational status, which was based on the highest value of maternal or paternal occupation (high-grade professional; low-grade professional; skilled worker/unskilled worker; student/economically inactive) adapted from the Danish International Classification of Occupations (DISCO-88) and the International Standard Classification of Education. Maternal age at birth and parity (primipara; multipara) were obtained from the Medical Birth Registry,⁵³ which contains information on all births in Denmark since 1973 and can be linked to the participants using the unique 10-digit civil registration (CPR) number assigned to all Danish citizens.

Characteristics among sons expected to be associated with male reproductive function were retrieved from the FEPOS questionnaire and the clinical examination.⁴³ These included BMI [measured by MC-780MA Body Composition Analyzer (Tanita®, Tokyo, Japan)], smoking (never; occasional or former smoker; current smoker), spillage of semen sample (no; yes), abstinence time in days (< 2 d, 2–4 d, ≥ 5 d), minutes from ejaculation to start of semen analysis, and time of day for blood sampling [morning (before 1200 hours), midday (1200–1800 hours), evening (after 1800 hours)].

Statistical Analyses

We initially examined the overall maternal PFAS plasma levels in the FEPOS cohort through calculation of medians and percentiles. To comply with the general data protection regulation (EU Regulation EU 2016/679 of 25 May 2018),⁵⁴ we calculated pseudo ranges, medians, and other percentiles based on information from at least five individuals who had values closest to the actual median/percentile. The highest median concentration in maternal plasma was that of PFOS and, therefore, PFOS was used to examine basic cohort characteristics across PFAS exposure tertiles. We then examined correlation patterns (Spearman's ρ) between maternal concentrations of single PFAS, as well as correlation patterns between maternal PFAS concentrations and PFAS concentrations, in adult sons. Next, we applied weighted quantile sum (WQS) regression⁵⁵ to estimate associations between combined exposure to maternal PFAS and measures of semen quality, testicular volume, and reproductive hormones. For the WQS regression, concentrations of the individual PFAS were divided into quartiles before creating a WQS index where each PFAS was assigned a weight (between 0 and 1 and summing up to 1). The weight reflected the specific strength of that PFAS in relation to the outcome while considering the collinearity between PFAS in the mixture. Using the gWQS package in R, data were split randomly into a training (40%) and a validation (60%) data set.^{42,55} The seed of R's random number generator was set to the date and time when we started to perform the analyses, to be able to create reproducible random data sets to be used in WQS regression. Empirical weights were estimated in the training data set, after which the association between the resulting WQS index and the outcome was tested in the validation data set. We restricted to 200 bootstrapped samples in all analyses, given that the results were robust at that point. A selection threshold equal to the inverse of the number of included PFAS ($1/n$, $n = 7$, because seven PFAS were included in the analyses) was applied to discriminate influential PFAS contributors to the overall association.⁴² Because WQS indices change a little each time they are constructed, we also considered contributions from PFAS 2% below the selection threshold. The WQS analyses allowed us to identify the overall combined associations for the PFAS on each reproductive outcome, as well as the contribution of the individual PFAS to the overall association. We *a priori* constrained (a necessary requirement for WQS analyses) associations for all mixture components to be negative, except for the association between the PFAS mixture and nonprogressive and immotile sperm, which was restricted to be positive.⁴² The WQS regression was performed with a Gaussian distribution assumption, because a negative binomial distribution ended in no convergence. The resulting WQS indices were afterward analyzed by negative binomial regression. Reported estimates and 95% confidence intervals (CIs) from the WQS analyses were presented as percentage differences of the outcome by a 1-unit increase in the WQS index. To test the assumption that the associations between single PFAS and reproductive function all had the same direction, and to compare results from the WQS analyses to analyses of single PFAS, we performed single-substance analyses. We used negative binomial regression to examine the association between maternal levels of each PFAS and reproductive function of the son. PFAS levels were analyzed both categorically in tertiles and continuously. We assessed the potential fit for several regression models, including Linear regression models, with or without transformations of independent and dependent variables, Poisson, and negative binomial models. Negative binomial regression models provided the best fit and were therefore used.

We present crude analyses for the total study population and adjusted analyses where relevant confounding factors and

precision variables were considered. The number of participants in adjusted analyses thus depended on the outcome variable and missing data in the covariates. Complete case analysis is presented in Tables S5 and S12–S15. Confounders were identified *a priori* based on existing literature and the use of a directed acyclic graph (Figure S1).⁵⁶ All analyses were adjusted for the following fetal risk factors: maternal prepregnancy BMI [underweight (< 18.5 kg/m²), normal weight (18.5–24.9 kg/m²), overweight (25–29.9 kg/m²), obese (> 30 kg/m²)], smoking (nonsmoker, 0–10 cigarettes/d, > 10 cigarettes/d), age at birth (continuous in years), and household occupational status (high-grade professional, low-grade professional, skilled/unskilled worker, student/economically inactive). Moreover, we adjusted for trimester of blood sampling (1, 2, 3),⁵⁷ and batch (1, 2). Participants reporting spillage during sample collection ($n = 149$) were excluded from the models examining semen volume and total sperm count. To increase precision, analyses of semen quality characteristics were further adjusted for spillage (not semen volume and total sperm count) and sexual abstinence time.⁵⁸ Analyses of motility were also adjusted for time from ejaculation to motility assessment. To increase precision in analyses of reproductive hormones, we adjusted for the time of day for blood sampling [morning (before 1200 hours), midday (1200–1800 hours), evening (after 1800 hours)] and sons' BMI [underweight (< 18.5 kg/m²), normal weight (18.5–24.9 kg/m²), overweight (25–29.9 kg/m²), obese (> 30 kg/m²)].⁵⁹

We performed two sensitivity analyses to examine the robustness of our results. First, we adjusted for the sons' plasma PFAS level to assess potential mediation of the association between maternal PFAS level and reproductive function. Second, we restricted to primipara, because children of primipara are exposed to higher levels of PFAS than children of later pregnancies given that concentrations are found to decrease with parity.^{19,60} We used a statistical significance level of 0.05 and used STATA-17 (StataCorp.) to calculate descriptive statistics and single-substance analyses and the gWQS package in R (version 4.1.0; R Development Core Team) to perform WQS regression analysis.

Ethical Approval

This study was conducted in accordance with the Declaration of Helsinki. The establishment of the FEPOS cohort was approved by the Scientific Research Ethics Committee for Copenhagen and Frederiksberg (no. H-16015857) and the Knowledge Center on Data Protection Compliance under the records of processing regarding health science research projects within the Capital Region of Denmark (no. 2012-58-0004). Recruitment and data collection were also permitted by the Steering Committee of the DNBC (reference nos. 2016-08 and 2019-21).

Results

Descriptive Analyses

The distribution of PFAS concentrations measured in maternal plasma are shown in Table 1. The highest median concentration was of PFOS [27.56 ng/mL; interquartile range (IQR): 12.74], followed by PFOA (4.40 ng/mL; IQR: 2.63). Maternal concentrations of PFOS and PFOA were highly correlated, as were PFDA, PFNA, and PFUnDA, respectively. For PFHpA, correlations with concentrations of other PFAS were poor (Spearman's $\rho \leq 0.33$; Table S3). We found negligible or weak correlations between maternal PFAS levels and PFAS levels in the sons (Table S4).⁴⁶

Baseline characteristics of the 864 included mothers and sons according to tertiles of maternal PFOS levels are presented in Table 2. Maternal PFOS levels ranged from 6.28 to 23.38 ng/mL in

Table 2. Baseline characteristics of the 864 sons, enrolled in the FEPOS cohort between 2017 and 2019 and their mothers, according to tertiles of maternal plasma PFOS concentrations.

Characteristics	Maternal PFOS level in tertiles			
	All <i>n</i> = 864	Low <i>n</i> = 288	Medium <i>n</i> = 288	High <i>n</i> = 288
Mother				
Age at birth (y) (mean ± SD)	31 ± 4	31 ± 4	30 ± 4	30 ± 4
Prepregnancy BMI (kg/m ²) [median (5th–95th P)] ^a	22 (18–30)	22 (18–28)	22 (19–30)	23 (18–31)
Smoking in first trimester [<i>n</i> (%)]				
Nonsmoker	668 (77)	217 (76)	222 (77)	229 (80)
Light smoker	168 (19)	64 (22)	56 (19)	48 (17)
Heavy smoker	28 (3)	7 (2)	10 (3)	11 (4)
Household occupational status [<i>n</i> (%)]				
High-grade profession	290 (34)	102 (35)	98 (34)	90 (31)
Low-grade profession	288 (33)	99 (34)	92 (32)	97 (34)
Skilled or unskilled worker	247 (29)	77 (27)	79 (27)	91 (32)
Student or economically inactive	39 (5)	10 (3)	19 (7)	10 (3)
Parity [<i>n</i> (%)]				
Primipara	374 (44)	94 (33)	127 (45)	153 (55)
Multipara	473 (56)	187 (67)	159 (55)	127 (45)
Son				
Age at clinical visit (y) (mean ± SD)	19 ± 0	19 ± 0	19 ± 0	19 ± 0
BMI (kg/m ²) [median (5th–95th P)] ^a	22 (18–28)	22 (18–28)	22 (18–27)	22 (18–29)
Smoking habits [<i>n</i> (%)]				
Never	405 (47)	123 (43)	132 (46)	150 (52)
Occasional or former smoker	245 (28)	75 (26)	79 (28)	91 (32)
Current smoker	211 (25)	89 (31)	76 (26)	46 (16)
Time of day for blood sampling [<i>n</i> (%)]				
Morning	310 (36)	100 (35)	100 (35)	110 (39)
Midday	454 (53)	156 (55)	152 (54)	146 (51)
Evening	91 (11)	30 (10)	32 (11)	29 (10)
Minutes from semen collection to analysis (mean ± SD)	50 ± 20	50 ± 19	52 ± 22	49 ± 19
Period of abstinence (d) [<i>n</i> (%)]				
<2	296 (35)	110 (39)	93 (32)	93 (33)
2–4	508 (60)	149 (53)	184 (64)	175 (62)
≥5	47 (6)	21 (8)	10 (3)	16 (6)
Spillage of semen sample [<i>n</i> (%)]				
Spillage	149 (17)	57 (20)	41 (14)	51 (18)
No spillage	707 (83)	227 (80)	246 (86)	234 (82)

Note: Tertiles are as follows: low (6.28–23.38 ng/mL), medium (23.51–31.49 ng/mL), high (31.66–92.59 ng/mL). Missing: Maternal age (*n* ≤ 5), prepregnancy BMI (*n* = 22), maternal smoking (*n* = 0), household occupational status (*n* = 0), parity (*n* = 17), age at clinical visit (*n* = 0), BMI (*n* ≤ 5), smoking habits (*n* ≤ 5), time of day for blood sampling (*n* = 9), minutes from semen collection to analysis (*n* = 11), days of abstinence (*n* ≤ 5), spillage (*n* = 8). BMI, body mass index; FEPOS, Fetal Programming of Semen Quality; P, percentile; PFOS, perfluorooctane sulfonic acid; SD, standard deviation.

^aPseudo median and percentiles were based on information from at least five individuals with values closest to the actual median/percentile.

the low-exposure tertile, whereas the medium- and high-exposure tertiles ranged from 23.51 to 31.49 ng/mL and 31.66 to 92.59 ng/mL, respectively. The mean age of the sons was 19 y, with an age range from 18.9 to 21.2 y. Sons within the high tertile of maternal exposure had mothers with a slightly lower household occupational status, were more often firstborn, less often smokers, and more often had an abstinence time of the recommended 2–4 d compared with low-exposed sons. Abstinence time was similar among sons in the medium and high tertiles of maternal exposure. Table 3 shows the pseudo medians and percentiles for semen quality characteristics, testicular volume, and levels of reproductive hormones according to PFOS exposure. Sons in the high tertile of PFOS exposure had lower sperm concentrations, total sperm counts, and percentage of progressive sperm cells and higher levels of calculated free testosterone compared with sons in the low tertile. There were no or less pronounced differences for the remaining semen quality characteristics, testicular volume, or reproductive hormones.

Combined PFAS Exposure

Crude and adjusted associations between exposure to maternal PFAS and reproductive function derived by the WQS regression analyses are shown in Table 4. In adjusted analyses, a 1-unit increase in the WQS index of combined maternal PFAS exposure was associated with lower sperm concentration (–8%; 95% CI:

–16%, –1%) and total sperm count (–10%; 95% CI: –17%, –2%), and a higher proportion of nonprogressive and immotile sperm (5%; 95% CI: 1%, 8%). We observed no association between the PFAS mixture and testicular volume or levels of reproductive hormones. Complete case analysis (*n* = 799) on the combined PFAS exposure did not differ from the analysis on the total cohort (Table S5). Figure 1 shows the WQS index weights for sperm concentration, total sperm count, and nonprogressive and immotile sperm. The PFAS contributing to the associations were PFHpA (42%), PFDA (19%), PFOS (17%), and PFUnDA (12%) for sperm concentration and PFHpA (52%), PFOA (16%), and PFHxS (12%) for total sperm count. For nonprogressive and immotile sperm, the contributors were PFHpA (65%), PFUnDA (15%), and PFOS (12%). The PFAS most heavily weighted in the indices was PFHpA in analyses of all three outcomes. Sensitivity analyses with adjustment for the sons' PFAS levels and primipara mothers did not change the main results substantially (Table S6). Associations between combined PFAS exposure and a lower percentage of morphologically normal sperm and a lower level of SHBG did, however, become significant when adjusting for sons' PFAS levels.

Single-Substance Analyses

In Table 5 the results from the single-substance analyses are presented for PFOS and PFHpA (the strongest contributor according

Table 3. Semen characteristics and reproductive hormones of the 864 sons enrolled in the FEPOS cohort between 2017 and 2019, according to tertiles of maternal plasma PFOS concentrations.

Semen characteristics and reproductive hormones	n	Maternal PFOS level in tertiles			
		All	Low	Medium	High
		Median (5th–95th P) ^a	Median (5th–95th P) ^a	Median (5th–95th P) ^a	Median (5th–95th P) ^a
Semen volume (mL) ^b	707	3 (1–5)	3 (1–6)	3 (1–5)	3 (1–5)
Sperm concentration ($\times 10^6$ /mL)	859	38 (3–136)	40 (3–136)	39 (2–135)	37 (4–138)
Total sperm count ($\times 10^6$) ^b	706	103 (8–406)	113 (9–405)	98 (8–398)	98 (6–425)
Progressive sperm (%)	847	64 (31–84)	68 (36–84)	64 (28–83)	62 (27–84)
Nonprogressive sperm (%)	847	7 (1–21)	6 (1–19)	7 (1–21)	8 (1–23)
Immotile sperm (%)	847	27 (13–55)	26 (12–52)	28 (14–61)	28 (13–56)
Morphologically normal sperm (%)	842	6 (0–15)	6 (0–15)	7 (1–15)	6 (0–15)
Average testicular volume (mL)	864	15 (7–25)	15 (7–25)	15 (7–25)	15 (8–25)
Follicle-stimulating hormone (IU/L)	854	4 (1–8)	4 (1–8)	4 (1–8)	3 (1–9)
Luteinizing hormone (IU/L)	854	5 (3–9)	5 (3–9)	5 (3–9)	5 (3–9)
Sexual hormone-binding globulin (nmol/L)	854	33 (17–59)	33 (17–58)	33 (19–59)	33 (16–59)
Estradiol (pmol/L)	854	53 (11–108)	53 (11–111)	54 (11–99)	52 (11–121)
Testosterone (nmol/L)	854	18 (10–29)	18 (10–29)	19 (10–29)	18 (11–30)
Calculated free testosterone (pmol/L)	854	385 (224–578)	382 (223–586)	387 (209–589)	387 (232–575)

Note: Tertiles are as follows: low (6.28–23.38 ng/mL), medium (23.51–31.49 ng/mL), high (31.66–92.59 ng/mL). FEPOS, Fetal Programming of Semen Quality; P, percentile; PFOS, perfluorooctane sulfonic acid.

^aPseudo median and percentiles were based on information from at least five individuals with values closest to the actual median/percentile.

^bSamples with spillage excluded ($n = 149$).

to the WQS indices). Total sperm count decreased by 5% (–5%; 95% CI: –9%, –1%) and proportion of nonprogressive and immotile sperm increased by 4% (4%; 95% CI: 2%, 7%) for each 10-ng/mL increase in PFOS. Sons in the high tertile of maternal PFHpA exposure had 13% (–13%; 95% CI: –24%, –1%) lower sperm concentration and 8% (8%; 95% CI: 1%, 15%) higher prevalence of nonprogressive and immotile sperm compared with sons in the low tertile. Tables S7–S11 present the results of the single-substance analyses of the other PFAS—all showing rather weak associations, with most confidence intervals including zero. We saw no clear associations between concentrations of single substances and testicular volume, or reproductive hormones (Tables 5 and S7–S11). Complete case analysis ($n = 799$) on the single PFAS did not differ from the analysis on the total cohort (Tables S12–S17).

Discussion

In this large-scale population-based study, both combined and single exposure to maternal PFAS concentrations during early pregnancy were associated with lower sperm concentration, lower total sperm count, and higher proportion of nonprogressive and immotile sperm in the adult sons. The WQS indices identified PFHpA, PFOA, PFOS, PFDA, PFHxS, and PFUnDA as contributors, with varying degrees of influence on the combined associations. PFHpA was the main contributor to all three outcomes. As expected, PFOS and PFOA were present in the highest concentrations in maternal plasma. Seven of the 15 measured PFAS were present above the LOD in >80% of the samples.

To our knowledge, only one previous epidemiological study has examined maternal PFAS exposure during pregnancy and semen quality in adulthood among offspring. Vested et al.³⁸ used multivariable regression analyses to study maternal exposure to PFOA and PFOS and found associations between higher plasma levels of PFOA and lower sperm concentration (–34%; 95% CI: –58, 5%), and total sperm count (–34%; 95% CI: –52, 12%) among 169 adult sons.³⁸ This is in line with our findings, although we did not observe as pronounced differences. In addition, we observed stronger associations between PFOS exposure and total sperm count and the proportion of nonprogressive and immotile sperm. Vested et al.³⁸ did not find a statistically significant association between PFOS and indicators of male reproductive function, which might be due to their smaller sample size compared with

our study. The maternal average levels of PFOA and PFOS in our study population are comparable to those reported by Vested et al.³⁸ and with several other previous studies on male reproductive function.¹⁸

Sensitivity analyses showed that restricting to primipara did not change the results substantially and that the sons' own PFAS

Table 4. Weighted quantile sum (WQS) regression analyses of reproductive parameters according to maternal plasma concentrations of the combined effect of PFAS among the 864 sons in the FEPOS cohort (measure of association is interpreted as percentage differences of the outcome by a 1-unit increase in the WQS index).

Reproductive parameters	Model	n	% (95% CI)
Semen volume (mL) ^a	Crude	707	–4 (–8, 1)
	Adjusted ^{b,c}	689	–4 (–9, 1)
Sperm concentration ($\times 10^6$ /mL)	Crude	858	–7 (–15, 1)
	Adjusted ^b	829	–8 (–16, –1)
Total sperm count ($\times 10^6$) ^a	Crude	706	–10 (–18, –2)
	Adjusted ^{b,c}	688	–10 (–17, –2)
Nonprogressive and immotile sperm (%)	Crude	847	4 (1, 8)
	Adjusted ^{b,c}	813	5 (1, 8)
Morphologically normal sperm (%)	Crude	842	–4 (–10, 2)
	Adjusted ^b	813	–6 (–12, 1)
Average testicular volume (mL)	Crude	864	–2 (–5, 1)
	Adjusted ^{b,c}	836	–2 (–5, 1)
Testosterone (nmol/L)	Crude	854	–1 (–4, 1)
	Adjusted ^c	829	–2 (–4, 1)
Estradiol (pmol/L)	Crude	854	–6 (–10, –1)
	Adjusted ^c	829	–3 (–8, 2)
Follicle-stimulating hormone (IU/L)	Crude	854	0 (–5, 4)
	Adjusted ^c	829	–1 (–5, 3)
Luteinizing hormone (IU/L)	Crude	854	3 (–1, 7)
	Adjusted ^c	829	3 (–1, 8)
Sexual hormone-binding globulin (nmol/L)	Crude	854	–2 (–6, 1)
	Adjusted ^c	829	–3 (–7, 0)
Calculated free testosterone (pmol/L)	Crude	854	–1 (–4, 2)
	Adjusted ^c	829	–1 (–3, 2)

Note: The number of participants in adjusted analyses depended on the outcome variable and missing data in the covariates. Values below the LOD were assigned the value of the LOD divided by the square root of 2. BMI, body mass index; CI, confidence interval; FEPOS, Fetal Programming of Semen Quality; LOD, limit of detection.

^aSamples with spillage excluded.

^bAdjusted for prepregnancy BMI, maternal age, household occupational status, maternal smoking, trimester of blood sampling, batch, abstinence time, and spillage.

^cNot adjusted for spillage.

^dFurther adjusted for time from ejaculation to analysis.

^eAdjusted for prepregnancy BMI, maternal age, household occupational status, maternal smoking, trimester of blood sampling, batch, sons BMI, time of day for blood sampling.

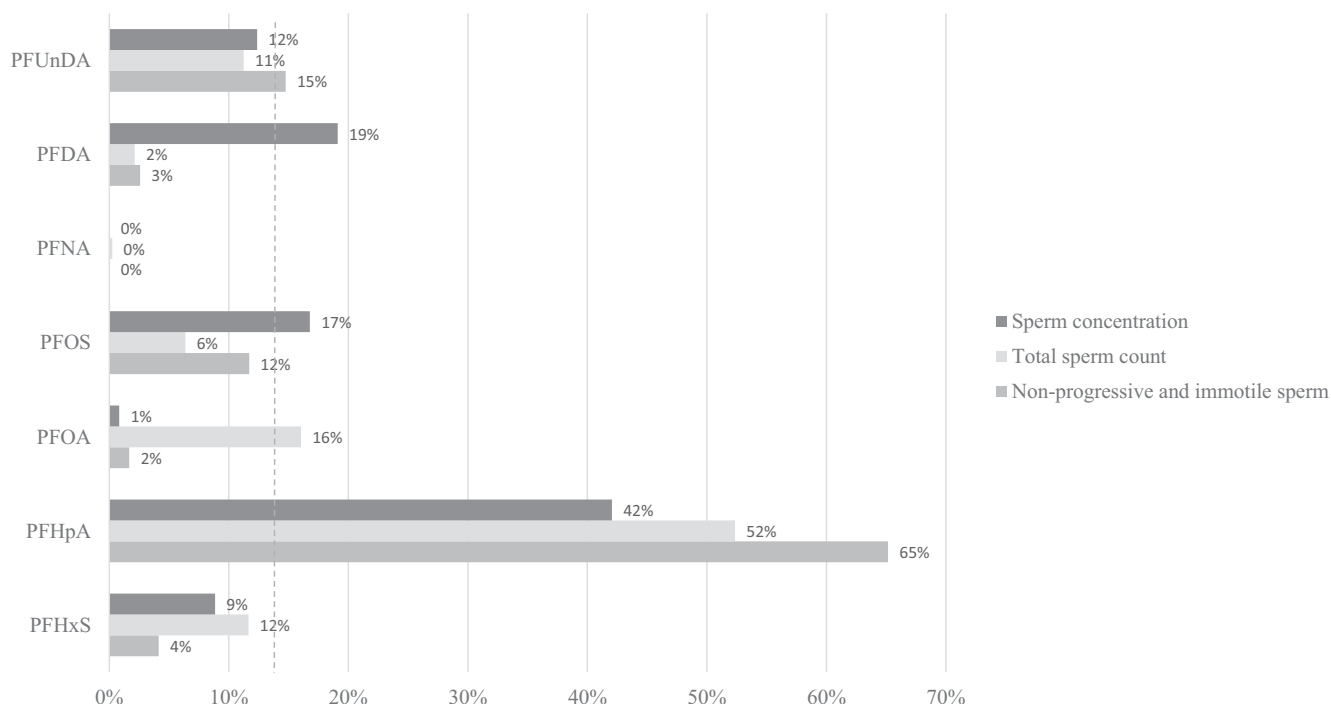


Figure 1. Weighted quantile sum (WQS) regression index weights in relation to sperm concentration, total sperm count, and nonprogressive and immotile sperm identifying contributions to the overall association. The dotted line represents the selection threshold of 14%, which is a guide to determine contributors. Values below LODs were assigned the value of the LOD divided by the square root of 2. Note: LOD, limit of detection; PFDA, perfluorodecanoic acid; PFNA, perfluorononanoic acid; PFHxS, perfluorohexane sulfonic acid; PFHpA, perfluoroheptanoic acid; PFOA, perfluorooctanoic acid; PFOS, perfluorooctane sulfonic acid; PFUnDA, perfluoroundecanoic acid.

exposure did not mediate the association, possibly owing to their relatively low levels⁴⁶ and weak correlation between levels of maternal PFAS and sons' PFAS. This is in line with our recent systematic review on PFAS exposure and male reproduction, where we concluded that there was no consistent indication of association between adult exposure to PFAS and semen quality.¹⁸

The potential for developmental toxicity, focusing on especially PFOA and PFOS, has been observed in several rodent studies, but the specific underlying mechanisms awaits clarification.^{61,62} A recent review on PFAS and male reproductive function describes that findings in rodent and *in vitro* studies support that PFOA and PFOS have potential for disruption of the male reproductive endocrine axis,⁶² given that associations between PFAS exposure and lower serum or testicular testosterone levels, inhibition of key steroidogenic enzymes activity, which might be due to the loss of Leydig cell function, and reduction in relevant testicular receptors have been observed.^{61–65} There are also indications of the ability of PFAS to disrupt the blood–testis barrier.^{62,66} Two previous studies found associations between fetal exposure to PFOS and increased levels of testosterone in amniotic fluid and estradiol in male infants.^{39,40} Our findings did not, however, indicate that exposure to maternal PFAS was associated with the adult sons' reproductive hormonal levels. However, if PFAS disrupts the hormonal environment during development of the male reproductive system, this might not manifest in changed levels of reproductive hormones in adulthood if the damage is not persistent.²⁶

According to the WQS indices, different PFAS contributed to the overall associations; however, PFHpA seemed to be the strongest contributor. Similar adverse associations with sperm concentration and proportion of nonprogressive and immotile sperm were observed in the single-substance analyses of PFHpA. If the associations are causal, PFHpA might have high placental transfer⁶⁷ and could be a highly potent human male reproductive toxicant.

According to *in vitro* and *in vivo* studies, several mechanisms, based on activation or inhibition of a range of receptors, may cause the reproductive toxicity of PFAS.⁶² PFAS carboxylic acids with a mid-chain length, such as PFHpA and PFOA, may have a high potential for activation of the peroxisome proliferator-activated receptors (PPARs), such as PPAR α .^{9,68} Although our knowledge of specific responses to receptor interaction remains incomplete, several PPARs appear to regulate Sertoli cell metabolism,⁶⁹ which could affect testicular functions.⁷⁰ However, with the relatively low concentrations and narrow exposure ranges present in our study, the observed strong contribution from PFHpA may at least, in part, be a chance finding. Given that testing multiple associations in the same data increase the risk of generating significant findings solely by coincidence, our findings should be interpreted with caution and weighed against existing evidence.

There are many ways to address combined effects. The WQS regression allows for investigation of several chemicals simultaneously while accounting for collinearity and mixture effects,⁴² and the model can estimate a joint effect of the entire mixture expressed as the effect of increasing all exposures by a single unit. A limitation of the WQS regression is the underlying assumption that the included exposures have a similar direction of effect (positive or negative).⁴² The single-substance analyses showed that not all PFAS were unambiguously associated with the adverse effects on semen quality that our WQS analyses were restricted to. Hence, if some of the PFAS exert a positive effect on semen quality, that would not be possible to identify. However, PFAS have not been described as beneficial for reproductive function.⁶² We used single imputation when values were below the LOD (LOD divided by the square root of 2). This was only relevant for PFHpA, where 11.5% ($n = 99$) of the values were below the LOD. Generally multiple imputation is recommended when there are <10% missing values,⁷¹ but because

Table 5. Negative binomial regression analyses of reproductive parameters according to maternal plasma PFOS and PFHpA concentrations among the 864 sons in the FEPOS cohort, with the lowest tertile as the reference group.

Reproductive parameters	Model	<i>n</i>	PFOS			PFHpA		
			Continuous per 10 ng/mL	Tertiles		Continuous per 0.1 ng/mL	Tertiles	
				Medium	High		Medium	High
			% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)
Semen volume (mL) ^a	Crude	711	-2 (-5, 1)	-3 (-11, 5)	-2 (-11, 7)	-1 (-4, 3)	-1 (-9, 8)	-1 (-9, 8)
	Adjusted ^{b,c}	693	-2 (-5, 1)	-3 (-11, 5)	-2 (-10, 7)	-1 (-4, 3)	0 (-8, 8)	0 (-8, 8)
Sperm concentration (× 10 ⁶ /mL)	Crude	859	-1 (-6, 4)	-4 (-14, 11)	-2 (-15, 13)	0 (-5, 5)	-10 (-21, 3)	-10 (-22, 4)
	Adjusted ^b	830	-2 (-7, 2)	-1 (-14, 13)	-5 (-17, 9)	-1 (-6, 4)	-10 (-22, 2)	-13 (-24, -1)
Total sperm count (× 10 ⁶) ^a	Crude	706	-1 (-7, 4)	-3 (-17, 14)	-1 (-16, 17)	-2 (-8, 4)	-8 (-23, 8)	-15 (-27, 0)
	Adjusted ^{b,c}	688	-5 (-9, -1)	-6 (-19, 8)	-12 (-24, 2)	-1 (-7, 4)	-6 (-19, 9)	-10 (-21, 4)
Nonprogressive and immotile sperm (%)	Crude	847	5 (2, 7)	9 (1, 16)	14 (6, 21)	2 (-1, 5)	3 (-4, 10)	6 (0, 14)
	Adjusted ^{b,d}	813	4 (2, 7)	8 (1, 17)	13 (5, 21)	2 (-1, 5)	4 (-3, 12)	8 (1, 15)
Morphologically normal sperm (%)	Crude	842	-1 (-5, 3)	8 (-3, 20)	-1 (-12, 11)	-1 (-6, 3)	-9 (-18, 2)	-6 (-16, 5)
	Adjusted ^b	813	-1 (-5, 2)	8 (-3, 21)	-2 (-13, 10)	-2 (-6, 3)	-11 (-21, 0)	-8 (-18, 2)
Average testicular volume (mL)	Crude	864	-1 (-2, 1)	-2 (-7, 3)	-3 (-8, 3)	0 (-2, 2)	-1 (-6, 5)	-1 (-6, 4)
	Adjusted ^{b,c}	836	-1 (-3, 1)	-3 (-8, 3)	-4 (-10, 1)	0 (-2, 2)	1 (-5, 6)	0 (-5, 6)
Testosterone (nmol/L)	Crude	854	-1 (-2, 1)	0 (-5, 6)	1 (-4, 7)	0 (-2, 2)	-2 (-7, 3)	-2 (-6, 4)
	Adjusted ^e	829	-1 (-2, 1)	0 (-5, 5)	1 (-4, 7)	-1 (-3, 1)	-3 (-8, 2)	-2 (-6, 3)
Estradiol (pmol/L)	Crude	854	-1 (-4, 2)	-4 (-11, 4)	-1 (-10, 9)	-1 (-5, 3)	-4 (-12, 5)	0 (-8, 9)
	Adjusted ^e	826	0 (-3, 3)	-4 (-11, 5)	1 (-8, 11)	-1 (-5, 3)	-6 (-14, 3)	-2 (-10, 7)
Follicle-stimulating hormone (IU/L)	Crude	854	-1 (-4, 3)	-5 (-14, 6)	-1 (-11, 10)	-2 (-6, 2)	0 (-10, 11)	-4 (-14, 7)
	Adjusted ^e	829	0 (-3, 4)	-4 (-13, 7)	1 (-10, 13)	-2 (-6, 3)	-1 (-12, 11)	-4 (-14, 7)
Luteinizing hormone (IU/L)	Crude	854	2 (0, 4)	6 (-1, 14)	7 (0, 14)	-1 (-4, 2)	-2 (-8, 5)	-3 (-9, 4)
	Adjusted ^e	829	2 (0, 4)	0 (0, 0)	0 (0, 0)	-1 (-4, 2)	-3 (-9, 5)	-3 (-9, 3)
Sexual hormone-binding globulin (nmol/L)	Crude	854	-1 (-4, 1)	1 (-5, 7)	0 (-6, 6)	1 (-2, 3)	2 (-4, 8)	1 (-5, 7)
	Adjusted ^e	829	-2 (-4, 1)	1 (-5, 7)	-1 (-7, 6)	0 (-2, 3)	2 (-4, 8)	0 (-5, 6)
Calculated free testosterone (pmol/L)	Crude	854	0 (-2, 2)	0 (-4, 5)	1 (-3, 6)	-1 (-3, 1)	-3 (-8, 1)	-2 (-6, 2)
	Adjusted ^e	829	0 (-2, 2)	0 (-5, 5)	2 (-3, 6)	-1 (-3, 1)	-4 (-8, 1)	-2 (-6, 2)

Note: The number of participants in adjusted analyses depended on the outcome variable and missing data in the covariates. Values below LODs were assigned the value of the LOD divided by the square root of 2. BMI, body mass index; CI, confidence interval; FEPOS, Fetal Programming of Semen Quality; LOD, limit of detection; PFHpA, perfluorheptanoic acid; PFOS, perfluorooctane sulfonic acid.

^aSamples with spillage excluded.

^bAdjusted for pre-pregnancy BMI, maternal age, household occupational status, maternal smoking, trimester of blood sampling, batch, abstinence time, and spillage.

^cNot adjusted for spillage.

^dFurther adjusted for time from ejaculation to analysis.

^eAdjusted for pre-pregnancy BMI, maternal age, household occupational status, maternal smoking, trimester of blood sampling, batch, sons BMI, time of day for blood sampling.

PFAS levels were divided into quartiles in the WQS analysis and tertiles in the analysis of single substances and imputed values therefore end up in the lowest quartile/tertile whether using single or multiple imputation, we believed this to give rise to negligible bias. All other PFAS included in analyses had all measurements above the LOD.

We expected minimal misclassification of maternal PFAS exposure given that it is measured with a highly sensitive LC-MS/MS technique¹⁹ by a HBM4EU-qualified laboratory undergoing interlaboratory control for analysis of PFAS.^{72,73} However, we were not able to measure PFAS concentrations in the target tissue in the fetus, and concentrations there may differ from those measured in maternal plasma.^{18,34,74} A newly published study demonstrated that placental transfer of PFAS is closely related to carbon chain length given that the binding affinities of PFAS to human serum albumin increased with chain length.⁷⁵ This means that maternal plasma levels of long-chain PFAS may be a better proxy of fetal exposure than short-chain PFAS. Most PFAS in this study were long-chained. We assumed blood samples drawn during early pregnancy to be a good proxy of the exposure during gestational weeks 7–15, where the gonadal development takes place.^{37,76} Our seven included PFAS have half-lives in humans ranging from 0.17 to 8.5 y,^{1,8,45,77} and maternal blood samples were drawn during the first trimester in 95% of the cases. Thus, changes in body composition, transfer to the fetus, and increased renal clearance later in pregnancy, which might reduce PFAS plasma levels,⁵⁷ are unlikely to have affected our measures substantially. To our knowledge, this study is the first to measure PFAS exposure in the beginning of pregnancy in relation to male reproductive

function, given that previous studies measured PFAS in mid- or late pregnancy^{38–40} or at birth, as a proxy of fetal exposure.⁶³

The participants in the FEPOS cohort delivered one semen and one blood sample that were analyzed according to standardized state-of-the-art techniques.⁴⁷ However, semen quality characteristics show intraindividual variability⁷⁸ and secretion of many reproductive hormones is pulsatile, whereby plasma concentrations vary across the day. Although repeated sampling could have reduced measurement error due to variability, asking for more than one sample may have reduced participation⁷⁹ and caused selection bias. A single semen sample may be sufficient for studies aimed at identifying average differences in semen quality across groups.⁸⁰ We tried to further increase precision of measures of both semen quality and reproductive hormones by controlling for several precision variables, such as abstinence time and time of day for blood sampling. We *a priori* included several potential confounders; however, residual confounding or confounding by other unknown factors, cannot be ruled out. There might be factors, such as other chemicals or lifestyle, that might be associated with PFAS exposure which also have an impact on reproductive function. However, we are not aware of potential confounding factors that are not accounted for in our analyses. Participants reporting spillage of their semen sample were excluded from the models examining semen volume and total sperm count and adjusted for in models examining the other sperm outcomes, because there were only minor differences between sons who reported spillage and sons who did not.

Groups with lower socioeconomic status in terms of education, occupation, income, and civil status are underrepresented in

the DNBC compared with the background population⁸¹ and thus also in the FEPOS cohort. The overall participation rate in the FEPOS cohort was relatively low (19%), and although participation rates <30% are not uncommon in studies of semen quality,⁸² selection bias could be of concern.^{78,82} However, baseline characteristics of participants, nonparticipants, and nonparticipants because of mothers with an insufficient amount of plasma available were comparable across relevant demographics, such as household occupational status, prepregnancy BMI, and maternal age at birth. Further, men trying to become fathers or worried about or diagnosed with fertility problems may have a higher propensity for participation in studies investigating semen quality.^{83,84} In our study, participants were young and unlikely to be worried about or aware of their fertility status. To cause selection bias, the likelihood of participation must furthermore depend on the exposure. Given that the sons were unaware of their mothers' PFAS levels during pregnancy, it is unlikely that the sons' participation depended on exposure status.

The main strength of this study was the very large sample size compared with previous studies^{38–40} with information on important aspects of factors in both fetal and adult life. Moreover, the FEPOS cohort provided a unique opportunity to combine extensive information from questionnaires and biological measurements provided by both mother and son and register data.

In conclusion, we find that combined maternal exposure to seven PFAS was associated with lower sperm concentration, lower total sperm count, and higher proportions of nonprogressive and immotile sperm in young adulthood. The WQS indices identified PFHpA, PFOA, PFOS, PFDA, PFHxS, and PFUnDA as contributors with varying degrees of influence on the combined associations. PFHpA was identified as the main contributor in relation to all three outcomes; however, cautious interpretation is needed given its low concentrations in maternal plasma and narrow range. Results from the single-substance analyses supported the findings of the combined exposure analysis. Thus, male reproductive toxicity of PFAS may extend beyond PFOA and PFOS, which have been the primary focus in previous studies. Although restrictions on the use of PFAS in the EU and other parts of the world have intensified, there are still areas with high concentrations. Further studies on maternal PFAS exposure and reproductive function, taking mixture effects and both legacy and emerging PFAS into account, are warranted.

Acknowledgments

C.H.R.H., B.B.H., A.G., J.P.B., and S.S.T. acquired funding. All authors made substantial contributions to conception and design of the study. Data was acquired by K.K.H., B.B.H., A.G., C.L., C.H.R.H., G.T., J.P.B., and S.S.T.. Analyses were performed by K.K.H., K.U.P., and E.M.F. The original draft was prepared by K.K.H. All authors contributed to interpretation of results, revision of the article, gave final approval of the version to be published, and agreed to be accountable for all aspects of the work.

We are grateful to all participants, to P.J. Clemmensen for her effort to get a grant for analyses of reproductive hormones, and to biomedical laboratory technicians M.L. Flensburg and J. Dideriksen for running the clinics and collecting data. We also thank J.R. Larsen for assisting with recruitment and data entry, and C. Tingsmark for conducting the morphology analysis. Moreover, we wish to thank A. Rönnholm, M. Bengtsson, and Å. Amilon at the Division of Occupational and Environmental Medicine at Lund University, Sweden, for performing the analyses of chemicals and the Department of Clinical Biochemistry at Aarhus University Hospital, Denmark, for conducting the hormone analyses.

The Danish National Birth Cohort was established with a significant grant from the Danish National Research Foundation. Additional support was obtained from the Danish Regional Committees, the Pharmacy Foundation, the Egmont Foundation, the March of Dimes Birth Defects Foundation, Helsefonden and other minor grants. The Danish National Birth Cohort Biobank has been supported by the Novo Nordisk Foundation and the Lundbeck Foundation. Follow-up of mothers and children have been supported by the Danish Medical Research Council (SSVF 0646, 271-08-0839/06-066023, O602-01042B, 0602-02738B), the Lundbeck Foundation (195/04, R100-A9193), The Innovation Fund Denmark 0603-00294B (09-067124), the Nordea Foundation (02-2013-2014), Aarhus Ideas (AU R9-A959-13-S804), University of Copenhagen Strategic Grant (IFSV 2012), and the Danish Council for Independent Research (DFF-4183-00594 and DFF-4183-00152).

The Fetal Programming of Semen Quality cohort and this study is part of the ReproUnion collaboration and co-financed by the European Union, Interreg V ÖKS (Öresund - Kattegat - Skagerrak), the Lundbeck Foundation, the Capital Region of Denmark, Medical Doctor Sofus Carl Emil Friis and Spouse Olga Doris Friis's Grant, Axel Muusfeldt's Foundation, A.P. Møller Foundation, Dagmar Marshalls Foundation, and Helsefonden. Karin Sørig Hougaard's contribution was supported by Focused Research Effort on Chemicals in the Working Environment (FFIKA), from the Danish Government.

References

- Ojo AF, Peng C, Ng JC. 2021. Assessing the human health risks of per- and polyfluoroalkyl substances: a need for greater focus on their interactions as mixtures. *J Hazard Mater* 407(Apr):124863, PMID: 33373965, <https://doi.org/10.1016/j.jhazmat.2020.124863>.
- Bonato M, Corrà F, Bellio M, Guidolin L, Tallandini L, Irato P, et al. 2020. PFAS environmental pollution and antioxidant responses: an overview of the impact on human field. *Int J Environ Res Public Health* 17(21):8020, PMID: 33143342, <https://doi.org/10.3390/ijerph17218020>.
- Kissa E. 2001. *Fluorinated Surfactants and Repellents*. 2nd ed. New York, NY: Marcel Dekker.
- IARC (International Agency for Research on Cancer). 2017. *Some Chemicals Used as Solvents and in Polymer Manufacture*. https://publications.iarc.fr/_publications/media/download/5626/38eb12059ccc7026d9c3b073e0ca7a7c667bd4c6.pdf [accessed 2 June 2021].
- Kotthoff M, Müller J, Jüriling H, Schlummer M, Fiedler D. 2015. Perfluoroalkyl and polyfluoroalkyl substances in consumer products. *Environ Sci Pollut Res Int* 22(19):14546–14559, PMID: 25854201, <https://doi.org/10.1007/s11356-015-4202-7>.
- Butenhoff JL, Olsen GW, Pfahles-Hutchens A. 2006. The applicability of biomonitoring data for perfluorooctanesulfonate to the environmental public health continuum. *Environ Health Perspect* 114(11):1776–1782, PMID: 17107867, <https://doi.org/10.1289/ehp.9060>.
- Kantiani L, Llorca M, Sanchis J, Farré M, Barceló D. 2010. Emerging food contaminants: a review. *Anal Bioanal Chem* 398(6):2413–2427, PMID: 20680618, <https://doi.org/10.1007/s00216-010-3944-9>.
- Olsen GW, Burris JM, Ehresman DJ, Froehlich JW, Seacat AM, Butenhoff JL, et al. 2007. Half-life of serum elimination of perfluorooctanesulfonate, perfluorohexanesulfonate, and perfluorooctanoate in retired fluorocarbon production workers. *Environ Health Perspect* 115(9):1298–1305, PMID: 17805419, <https://doi.org/10.1289/ehp.10009>.
- Danish Environmental Protection Agency. 2015. *Short-Chain Polyfluoroalkyl Substances (PFAS)—A Literature Review of Information on Human Health Effects and Environmental Fate and Effect Aspects of Short-Chain PFAS*. Environmental Project No. 1707. <https://www2.mst.dk/Udgiv/publications/2015/05/978-87-93352-15-5.pdf> [accessed 12 February 2021].
- European Commission. 2019. Regulation (EU) 2019/1021 of the European Parliament and of the Council of 20 June 2019 on persistent organic pollutants (recast). *Off J Eur Union L* 169:45–77. <https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX:32019R1021> [accessed 20 July 2021].
- Mamsen LS, Björvang RD, Mucs D, Vinnars MT, Papadogiannakis N, Lindh CH, et al. 2019. Concentrations of perfluoroalkyl substances (PFASs) in human embryonic and fetal organs from first, second, and third trimester pregnancies. *Environ Int* 124:482–492, PMID: 30684806, <https://doi.org/10.1016/j.envint.2019.01.010>.

12. Bach CC, Bech BH, Brix N, Nohr EA, Bonde JPE, Henriksen TB. 2015. Perfluoroalkyl and polyfluoroalkyl substances and human fetal growth: a systematic review. *Crit Rev Toxicol* 45(1):53–67, PMID: 25372700, <https://doi.org/10.3109/10408444.2014.952400>.
13. Tsuda S. 2016. Differential toxicity between perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA). *J Toxicol Sci* 41(special):SP27–SP36, PMID: 28003637, <https://doi.org/10.2131/jts.41.SP27>.
14. Wang Z, Dewitt JC, Higgins CP, Cousins IT. 2017. A never-ending story of per- and polyfluoroalkyl substances (PFASs)? *Environ Sci Technol* 51(5):2508–2518, PMID: 28224793, <https://doi.org/10.1021/acs.est.6b04806>.
15. Bjerregaard-Olesen C, Bach CC, Long M, Ghisari M, Bech BH, Nohr EA, et al. 2016. Determinants of serum levels of perfluorinated alkyl acids in Danish pregnant women. *Int J Hyg Environ Health* 219(8):867–875, PMID: 27451073, <https://doi.org/10.1016/j.ijheh.2016.07.008>.
16. Brase RA, Mullin EJ, Spink DC. 2021. Legacy and emerging per- and polyfluoroalkyl substances: analytical techniques, environmental fate, and health effects. *Int J Mol Sci* 22(3):995, PMID: 33498193, <https://doi.org/10.3390/ijms22030995>.
17. Pan Y, Cui Q, Wang J, Sheng N, Jing J, Yao B, et al. 2019. Profiles of emerging and legacy per-/polyfluoroalkyl substances in matched serum and semen samples: new implications for human semen quality. *Environ Health Perspect* 127(12):127005, PMID: 31841032, <https://doi.org/10.1289/EHP4431>.
18. Petersen KU, Larsen JR, Deen L, Flachs EM, Hærvig KK, Hull SD, et al. 2020. Per- and polyfluoroalkyl substances and male reproductive health: a systematic review of the epidemiological evidence. *J Toxicol Environ Health B Crit Rev* 23(6):276–291, PMID: 32741292, <https://doi.org/10.1080/10937404.2020.1798315>.
19. Norén E, Lindh C, Glynn A, Rylander L, Pineda D, Nielsen C. 2021. Temporal trends, 2000–2017, of perfluoroalkyl acid (PFAA) concentrations in serum of Swedish adolescents. *Environ Int* 155(Oct):106716, PMID: 34144476, <https://doi.org/10.1016/j.envint.2021.106716>.
20. Bjerregaard-Olesen C, Bach CC, Long M, Wielsøe M, Bech BH, Henriksen TB, et al. 2019. Associations of fetal growth outcomes with measures of the combined xenoestrogenic activity of maternal serum perfluorinated alkyl acids in Danish pregnant women. *Environ Health Perspect* 127(1):017006, PMID: 30676078, <https://doi.org/10.1289/EHP1884>.
21. Shu H, Lindh CH, Wikström S, Bornehag CG. 2018. Temporal trends and predictors of perfluoroalkyl substances serum levels in Swedish pregnant women in the SELMA study. *PLoS One* 13(12):e0209255, PMID: 30596681, <https://doi.org/10.1371/journal.pone.0209255>.
22. Joensen UN, Bossi R, Leffers H, Jensen AA, Skakkebaek NE, Jørgensen N. 2009. Do perfluoroalkyl compounds impair human semen quality? *Environ Health Perspect* 117(6):923–927, PMID: 19590684, <https://doi.org/10.1289/ehp.0800517>.
23. Joensen UN, Veyrand B, Antignac JP, Blomberg Jensen M, Petersen JH, Marchand P, et al. 2013. PFOS (perfluorooctanesulfonate) in serum is negatively associated with testosterone levels, but not with semen quality, in healthy men. *Hum Reprod* 28(3):599–608, PMID: 23250927, <https://doi.org/10.1093/humrep/des425>.
24. Ingelido AM, Abballe A, Gemma S, Dellatte E, Iacovella N, De Angelis G, et al. 2018. Biomonitoring of perfluorinated compounds in adults exposed to contaminated drinking water in the Veneto Region, Italy. *Environ Int* 110(Jan):149–159, PMID: 29108835, <https://doi.org/10.1016/j.envint.2017.10.026>.
25. Bach CC, Vested A, Jørgensen KT, Bonde JP, Henriksen TB, Toft G. 2016. Perfluoroalkyl and polyfluoroalkyl substances and measures of human fertility: a systematic review. *Crit Rev Toxicol* 46(9):735–755, PMID: 27268162, <https://doi.org/10.1080/10408444.2016.1182117>.
26. Nian M, Luo K, Luo F, Aimuzi R, Huo X, Chen Q, et al. 2020. Association between prenatal exposure to PFAS and fetal sex hormones: are the short-chain PFAS safer? *Environ Sci Technol* 54(13):8291–8299, PMID: 32525661, <https://doi.org/10.1021/acs.est.0c02444>.
27. Lau C, Anitole K, Hodes C, Lai D, Pfahles-Hutchens A, Seed J. 2007. Perfluoroalkyl acids: a review of monitoring and toxicological findings. *Toxicol Sci* 99(2):366–394, PMID: 17519394, <https://doi.org/10.1093/toxsci/kfm128>.
28. Coperchini F, Croce L, Ricci G, Magri F, Rotondi M, Imbriani M, et al. 2021. Thyroid disrupting effects of old and new generation PFAS. *Front Endocrinol (Lausanne)* 11(Jan):612320, PMID: 33542707, <https://doi.org/10.3389/fendo.2020.612320>.
29. Steenland K, Winquist A. 2021. PFAS and cancer, a scoping review of the epidemiologic evidence. *Environ Res* 194(Mar):110690, PMID: 33385391, <https://doi.org/10.1016/j.envres.2020.110690>.
30. Blake BE, Fenton SE. 2020. Early life exposure to per- and polyfluoroalkyl substances (PFAS) and latent health outcomes: a review including the placenta as a target tissue and possible driver of peri- and postnatal effects. *Toxicology* 443(Oct):152565, PMID: 32861749, <https://doi.org/10.1016/j.tox.2020.152565>.
31. Bonde JP, Flachs EM, Rimborg S, Glazer CH, Giwercman A, Ramlau-Hansen CH, et al. 2016. The epidemiologic evidence linking prenatal and postnatal exposure to endocrine disrupting chemicals with male reproductive disorders: a systematic review and meta-analysis. *Hum Reprod Update* 23(1):104–125, PMID: 27655588, <https://doi.org/10.1093/humupd/dmw036>.
32. Perry MJ, Nguyen GN, Porter ND. 2016. *The Current Epidemiologic Evidence on Exposures to Poly- and Perfluoroalkyl Substances (PFASs) and Male Reproductive Health*. Cham, Switzerland: Springer International Publishing.
33. Mamsen LS, Jönsson BAG, Lindh CH, Olesen RH, Larsen A, Ernst E, et al. 2017. Concentration of perfluorinated compounds and cotinine in human foetal organs, placenta, and maternal plasma. *Sci Total Environ* 596–597:97–105, PMID: 28426990, <https://doi.org/10.1016/j.scitotenv.2017.04.058>.
34. Björvang RD, Vinnars MT, Papadogiannakis N, Gidlöf S, Mamsen LS, Mucs D, et al. 2021. Mixtures of persistent organic pollutants are found in vital organs of late gestation human fetuses. *Chemosphere* 283:131125, PMID: 34467953, <https://doi.org/10.1016/j.chemosphere.2021.131125>.
35. Vesterholm Jensen D, Christensen J, Virtanen HE, Skakkebaek NE, Main KM, Toppari J, et al. 2014. No association between exposure to perfluorinated compounds and congenital cryptorchidism: a nested case-control study among 215 boys from Denmark and Finland. *Reproduction* 147(4):411–417, PMID: 24218628, <https://doi.org/10.1530/REP-13-0444>.
36. Ernst A, Brix N, Lauridsen LLB, Olsen J, Parner ET, Liew Z, et al. 2019. Exposure to perfluoroalkyl substances during fetal life and pubertal development in boys and girls from the Danish National Birth Cohort. *Environ Health Perspect* 127(1):017004, PMID: 30628845, <https://doi.org/10.1289/EHP3567>.
37. Skakkebaek NE, Rajpert-De Meyts E, Buck Louis GM, Toppari J, Andersson AM, Eisenberg ML, et al. 2016. Male reproductive disorders and fertility trends: influences of environment and genetic susceptibility. *Physiol Rev* 96(1):55–97, PMID: 26582516, <https://doi.org/10.1152/physrev.00017.2015>.
38. Vested A, Ramlau-Hansen CH, Olsen SF, Bonde JP, Kristensen SL, Halldorsson TI, et al. 2013. Associations of *in utero* exposure to perfluorinated alkyl acids with human semen quality and reproductive hormones in adult men. *Environ Health Perspect* 121(4):453–458, PMID: 23360585, <https://doi.org/10.1289/ehp.1205118>.
39. Itoh S, Araki A, Mitsui T, Miyashita C, Goudarzi H, Sasaki S, et al. 2016. Association of perfluoroalkyl substances exposure in utero with reproductive hormone levels in cord blood in the Hokkaido Study on Environment and Children's Health. *Environ Int* 94(Sep):51–59, PMID: 27209000, <https://doi.org/10.1016/j.envint.2016.05.011>.
40. Toft G, Jönsson BAG, Bonde JP, Nørgaard-Pedersen B, Hougaard DM, Cohen A, et al. 2016. Perfluorooctane sulfonate concentrations in amniotic fluid, biomarkers of fetal Leydig cell function, and cryptorchidism and hypospadias in Danish boys (1980–1996). *Environ Health Perspect* 124(1):151–156, PMID: 26046833, <https://doi.org/10.1289/ehp.1409288>.
41. Sharpe RM. 2020. Androgens and the masculinization programming window: human–rodent differences. *Biochem Soc Trans* 48(4):1725–1735, PMID: 32779695, <https://doi.org/10.1042/BST20200200>.
42. Carrico C, Gennings C, Wheeler DC, Factor-Litvak P. 2015. Characterization of a weighted quantile score approach for highly correlated data in risk analysis scenarios. *J Agric Biol Environ Stat* 20(1):100–120, <https://doi.org/10.1007/s13253-014-0180-3>.
43. Keglberg Hærvig K, Bonde JP, Ramlau-Hansen CH, Toft G, Hougaard KS, Specht IO, et al. 2020. Fetal Programming of Semen Quality (FEPOS) cohort—a DNBC male-offspring cohort. *Clin Epidemiol* 12(Jul):757–770, PMID: 32765110, <https://doi.org/10.2147/CLEP.S242631>.
44. Olsen J, Melbye M, Olsen SF, Sørensen TI, Aaby P, Andersen AM, et al. 2001. The Danish National Birth Cohort—its background, structure and aim. *Scand J Public Health* 29(4):300–307, PMID: 11775787, <https://doi.org/10.1177/14034948010290040201>.
45. Xu Y, Fletcher T, Pineda D, Lindh CH, Nilsson C, Glynn A, et al. 2020. Serum half-lives for short- and long-chain perfluoroalkyl acids after ceasing exposure from drinking water contaminated by firefighting foam. *Environ Health Perspect* 128(7):077004, PMID: 32648786, <https://doi.org/10.1289/EHP6785>.
46. Petersen KU, Hærvig KK, Flachs EM, Bonde JP, Lindh C, Hougaard KS, et al. 2022. Per- and polyfluoroalkyl substances (PFAS) and male reproductive function in young adulthood: a cross-sectional study. *Environ Res* 212(pt A):113157, PMID: 35318009, <https://doi.org/10.1016/j.envres.2022.113157>.
47. WHO (World Health Organization). 2010. *WHO Laboratory Manual for the Examination and Processing of Human Semen, Fifth Edition*. https://www.who.int/docs/default-source/reproductive-health/srhr-documents/infertility/examination-and-processing-of-human-semen-5ed-eng.pdf?sfvrsn=5227886e_2 [accessed 25 August 2016].
48. Ramlau-Hansen CH, Thulstrup AM, Bonde JP, Ernst E. 2007. Is self-measuring of testicular volume by a Prader orchidometer a valid method? *Fertil Steril* 87(6):1480–1482, PMID: 17296192, <https://doi.org/10.1016/j.fertnstert.2006.11.032>.

49. Harwood DT, Handelsman DJ. 2009. Development and validation of a sensitive liquid chromatography–tandem mass spectrometry assay to simultaneously measure androgens and estrogens in serum without derivatization. *Clin Chim Acta* 409(1–2):78–84, PMID: [19747904](#), <https://doi.org/10.1016/j.cca.2009.09.003>.
50. Vermeulen A, Verdonck L, Kaufman JM. 1999. A critical evaluation of simple methods for the estimation of free testosterone in serum. *J Clin Endocrinol Metab* 84(10):3666–3672, PMID: [10523012](#), <https://doi.org/10.1210/jcem.84.10.6079>.
51. Ko K, Kwon MJ, Woo HY, Park H. 2015. Evaluation of analytical performance of the Cobas 8000 analyzer series module e602. *J Lab Med Qual Assur* 37(2):101–109, <https://doi.org/10.15263/jlmqa.2015.37.2.101>.
52. Nybo Andersen AM, Olsen J. 2011. The Danish National Birth Cohort: selected scientific contributions within perinatal epidemiology and future perspectives. *Scand J Public Health* 39(7 suppl):115–120, PMID: [21775368](#), <https://doi.org/10.1177/1403494811407674>.
53. Bliddal M, Broe A, Pottegård A, Olsen J, Langhoff-Roos J. 2018. The Danish Medical Birth Register. *Eur J Epidemiol* 33(1):27–36, PMID: [29349587](#), <https://doi.org/10.1007/s10654-018-0356-1>.
54. European Commission. 2016. Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data on the free movement of such data, and repealing Directive 95/46/ED (General Data Protection Regulation). *Off J Eur Union L* 119:1–99. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:32016R0679> [accessed 13 September 2022].
55. Tanner EM, Bornehag CG, Gennings C. 2019. Repeated holdout validation for weighted quantile sum regression. *MethodsX* 6(Nov):2855–2860, PMID: [31871919](#), <https://doi.org/10.1016/j.mex.2019.11.008>.
56. Greenland S, Pearl J, Robins JM. 1999. Causal diagrams for epidemiologic research. *Epidemiology* 10(1):37–48, PMID: [9888278](#), <https://doi.org/10.1097/00001648-199901000-00008>.
57. Nielsen C, Andersson Hall U, Lindh C, Ekström U, Xu Y, Li Y, et al. 2020. Pregnancy-induced changes in serum concentrations of perfluoroalkyl substances and the influence of kidney function. *Environ Health* 19(1):80, PMID: [32641055](#), <https://doi.org/10.1186/s12940-020-00626-6>.
58. Comar VA, Petersen CG, Mauri AL, Mattila M, Vagnini LD, Renzi A, et al. 2017. Influence of the abstinence period on human sperm quality: analysis of 2,458 semen samples. *JBRA Assist Reprod* 21(4):306–312, PMID: [28985041](#), <https://doi.org/10.5935/1518-0557.20170052>.
59. Glenn T, Harris AL, Lindheim SR. 2019. Impact of obesity on male and female reproductive outcomes. *Curr Opin Obstet Gynecol* 31(4):201–206, PMID: [31022079](#), <https://doi.org/10.1097/GCO.0000000000000549>.
60. Brantsæter AL, Whitworth KW, Ydersbond TA, Haug LS, Haugen M, Knutsen HK, et al. 2013. Determinants of plasma concentrations of perfluoroalkyl substances in pregnant Norwegian women. *Environ Int* 54(Apr):74–84, PMID: [23419425](#), <https://doi.org/10.1016/j.envint.2012.12.014>.
61. Eggert A, Cisneros-Montalvo S, Anandan S, Musilli S, Stukenborg JB, Adamsson A, et al. 2019. The effects of perfluorooctanoic acid (PFOA) on fetal and adult rat testis. *Reprod Toxicol* 90(Dec):68–76, PMID: [31412280](#), <https://doi.org/10.1016/j.reprotox.2019.08.005>.
62. Tarapore P, Ouyang B. 2021. Perfluoroalkyl chemicals and male reproductive health: do PFOA and PFOS increase risk for male infertility? *Int J Environ Res Public Health* 18(7):3794, PMID: [33916482](#), <https://doi.org/10.3390/ijerph18073794>.
63. Yao Q, Shi R, Wang C, Han W, Gao Y, Zhang Y, et al. 2019. Cord blood *per*- and *poly*fluoroalkyl substances, placental steroidogenic enzyme, and cord blood reproductive hormone. *Environ Int* 129(Aug):573–582, PMID: [31174145](#), <https://doi.org/10.1016/j.envint.2019.03.047>.
64. Chen J, Wang X, Ge X, Wang D, Wang T, Zhang L, et al. 2016. Chronic perfluorooctanesulphonic acid (PFOS) exposure produces estrogenic effects in zebrafish. *Environ Pollut* 218:702–708, PMID: [27496563](#), <https://doi.org/10.1016/j.envpol.2016.07.064>.
65. Chen L, Lam JCW, Hu C, Tsui MMP, Lam PKS, Zhou B. 2019. Perfluorobutanesulfonate exposure skews sex ratio in fish and transgenerationally impairs reproduction. *Environ Sci Technol* 53(14):8389–8397, PMID: [31269390](#), <https://doi.org/10.1021/acs.est.9b01711>.
66. Li N, Mruk DD, Chen H, Wong CKC, Lee WM, Cheng CY, et al. 2016. Rescue of perfluorooctanesulfonate (PFOS)-mediated Sertoli cell injury by overexpression of gap junction protein connexin 43. *Sci Rep* 6(Jul):29667, PMID: [27436542](#), <https://doi.org/10.1038/srep29667>.
67. Cai D, Li QQ, Chu C, Wang SZ, Tang YT, Appleton AA, et al. 2020. High trans-placental transfer of perfluoroalkyl substances alternatives in the matched maternal-cord blood serum: evidence from a birth cohort study. *Sci Total Environ* 705(Feb):135885, PMID: [31841927](#), <https://doi.org/10.1016/j.scitotenv.2019.135885>.
68. Buhrke T, Kibellus A, Lampen A. 2013. *In vitro* toxicological characterization of perfluorinated carboxylic acids with different carbon chain lengths. *Toxicol Lett* 218(2):97–104, PMID: [23391484](#), <https://doi.org/10.1016/j.toxlet.2013.01.025>.
69. Rouiller-Fabre V, Guerin MJ, N'Tumba-Byn T, Muczynski V, Moison D, Tourpin S, et al. 2015. Nuclear receptors and endocrine disruptors in fetal and neonatal testes: a gapped landscape. *Front Endocrinol (Lausanne)* 6(May):58, PMID: [25999913](#), <https://doi.org/10.3389/fendo.2015.00058>.
70. Monrose M, Thirouard L, Garcia M, Holota H, De Haze A, Caira F, et al. 2021. New perspectives on PPAR, VDR and RXRα as new actors in testicular pathophysiology. *Mol Aspects Med* 78(Apr):100886, PMID: [32878696](#), <https://doi.org/10.1016/j.mam.2020.100886>.
71. Hargarten PM, Wheeler DC. 2020. Accounting for the uncertainty due to chemicals below the detection limit in mixture analysis. *Environ Res* 186(Jul):109466, PMID: [32344207](#), <https://doi.org/10.1016/j.envres.2020.109466>.
72. Lindh CH, Rylander L, Toft G, Axmon A, Rignell-Hydbom A, Giwercman A, et al. 2012. Blood serum concentrations of perfluorinated compounds in men from Greenlandic Inuit and European populations. *Chemosphere* 88(11):1269–1275, PMID: [22494529](#), <https://doi.org/10.1016/j.chemosphere.2012.03.049>.
73. Gyllenhammar I, Glynn A, Jönsson BAG, Lindh CH, Darnerud PO, Svensson K, et al. 2017. Diverging temporal trends of human exposure to bisphenols and plasticizers, such as phthalates, caused by substitution of legacy EDCs? *Environ Res* 153:48–54, PMID: [27898309](#), <https://doi.org/10.1016/j.envres.2016.11.012>.
74. Ng CA, Hungerbühler K. 2013. Bioconcentration of perfluorinated alkyl acids: how important is specific binding? *Environ Sci Technol* 47(13):7214–7223, PMID: [23734664](#), <https://doi.org/10.1021/es400981a>.
75. Lu Y, Meng L, Ma D, Cao H, Liang Y, Liu H, et al. 2021. The occurrence of PFAS in human placenta and their binding abilities to human serum albumin and organic anion transporter 4. *Environ Pollut* 273(Jan):116460, PMID: [33485002](#), <https://doi.org/10.1016/j.envpol.2021.116460>.
76. Toft G, Jönsson BAG, Lindh CH, Giwercman A, Spano M, Heederik D, et al. 2012. Exposure to perfluorinated compounds and human semen quality in Arctic and European populations. *Hum Reprod* 27(8):2532–2540, PMID: [22647447](#), <https://doi.org/10.1093/humrep/des185>.
77. Zhang Y, Beesoon S, Zhu L, Martin JW. 2013. Biomonitoring of perfluoroalkyl acids in human urine and estimates of biological half-life. *Environ Sci Technol* 47(18):10619–10627, PMID: [23980546](#), <https://doi.org/10.1021/es401905e>.
78. Wilcox AJ. 2010. *Fertility and Pregnancy - An Epidemiologic Perspective*. New York, NY: Oxford University Press.
79. Stokes-Riner A, Thurston SW, Brazil C, Guzick D, Liu F, Overstreet JW, et al. 2007. One semen sample or 2? Insights From a study of fertile men. *J Androl* 28(5):638–643, PMID: [17522415](#), <https://doi.org/10.2164/jandrol.107.002741>.
80. Chiu YH, Edifor R, Rosner BA, Nassan FL, Gaskins AJ, Mínguez-Alarcón L, et al. 2017. What does a single semen sample tell you? Implications for male factor infertility research. *Am J Epidemiol* 186(8):918–926, PMID: [28541378](#), <https://doi.org/10.1093/aje/kwx169>.
81. Jacobsen TN, Nohr EA, Frydenberg M. 2010. Selection by socioeconomic factors into the Danish National Birth Cohort. *Eur J Epidemiol* 25(5):349–355, PMID: [20349116](#), <https://doi.org/10.1007/s10654-010-9448-2>.
82. Larsen SB, Abell A, Bonde JP. 1998. Selection bias in occupational sperm studies. *Am J Epidemiol* 147(7):681–685, PMID: [9554607](#), <https://doi.org/10.1093/oxfordjournals.aje.a009509>.
83. Eustache F, Auger J, Cabrol D, Jouannet P. 2004. Are volunteers delivering semen samples in fertility studies a biased population? *Hum Reprod* 19(12):2831–2837, PMID: [15459173](#), <https://doi.org/10.1093/humrep/deh503>.
84. Bonde JP, Giwercman A, Ernst E. 1996. Identifying environmental risk to male reproductive function by occupational sperm studies: logistics and design options. *Occup Environ Med* 53(8):511–519, PMID: [8983461](#), <https://doi.org/10.1136/oem.53.8.511>.